

CS 285 — HW1: Imitation Learning (Push-T)

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Summary

Part	Policy	Best eval reward	Threshold
2	MSE action chunking	0.636	0.5 ✓
3	Flow matching	0.845	0.7 ✓

Both policies use the same 3×256 ReLU MLP backbone, chunk size $K = 8$, Adam at $\text{lr } 3 \times 10^{-4}$, batch 128, 400 epochs. The flow policy widens the input to carry the noisy chunk and flow timestep, and integrates a learned velocity field at inference.

Part 2 — Action Chunking with MSE Loss

Result

The MSE policy reaches a **best eval mean reward of 0.636** (final 0.636), clearing ≥ 0.5 . Training loss falls from ≈ 1.0 to ≈ 0.016 .

Metric	Value
Best / final eval mean reward	0.636
Final train loss	0.016
Training steps	75,600 (400 epochs)

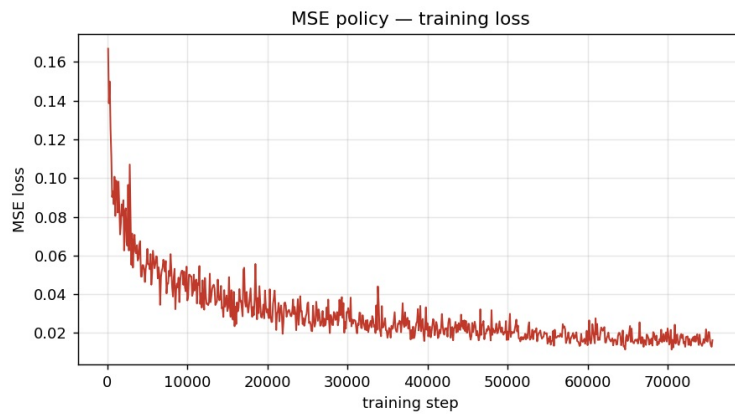
Architecture

A plain MLP mapping one state to a flat action chunk, reshaped to $(K, \text{action_dim})$.

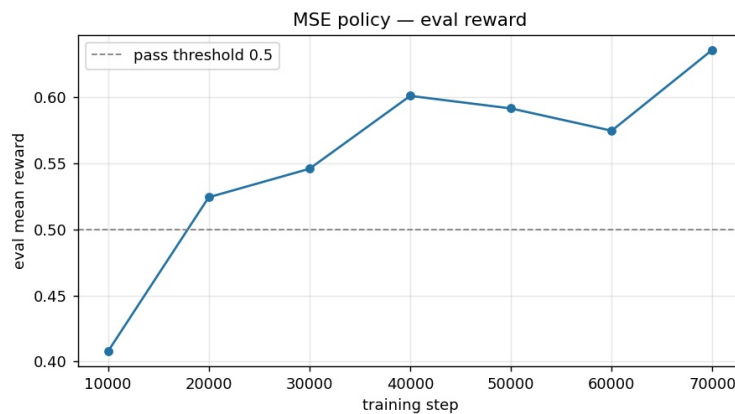
- **Input:** $\text{state_dim} = 5$ (agent x, y ; $T_{x,y}, \theta$)
- **Hidden:** 3×256 , ReLU after each
- **Output:** $8 \times 2 = 16$, reshaped to $(8, 2)$; no output activation (actions are raw coordinates)
- **Pipeline:** $5 \rightarrow 256 \rightarrow 256 \rightarrow 256 \rightarrow 16$

Loss: mean squared error between predicted and expert chunk (eq. 1).

Training curves



MSE training loss



MSE eval reward

Part 3 — Action Chunking with Flow Matching

Result

The flow policy reaches a **best eval mean reward of 0.845** (step 60k; final 0.834), clearing ≥ 0.7 — first crossing 0.7 by step 40k. The flow loss plateaus around **0.22**, not zero.

Metric	Value
Best eval mean reward	0.845 (step 60,000)
Final eval mean reward	0.834
Final train loss	0.220
Inference Euler steps n	10

Why the loss floors above zero (unlike MSE): the target velocity $A_t - A_{t,0}$ is stochastic — many (noise, data) pairs pass through one point $A_{t,\tau}$ with different velocities, so the net learns the conditional mean and an irreducible variance remains. Low flow loss is not the objective; eval reward is.

Architecture

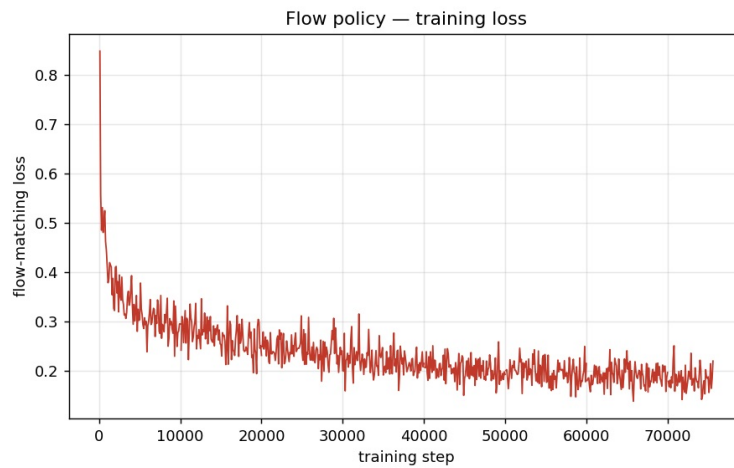
Same backbone, but the input carries the noisy chunk and timestep because the network is a **velocity field** $v_\theta(o_t, A_{t,\tau}, \tau)$, not a one-shot map.

- **Input:** state (5) + flattened noisy chunk (16) + τ (1) = 22
- **Hidden:** 3×256 , ReLU after each
- **Output:** $16 \rightarrow (8, 2)$ predicted velocity; no output activation (velocity is signed)
- **Pipeline:** $22 \rightarrow 256 \rightarrow 256 \rightarrow 256 \rightarrow 16$

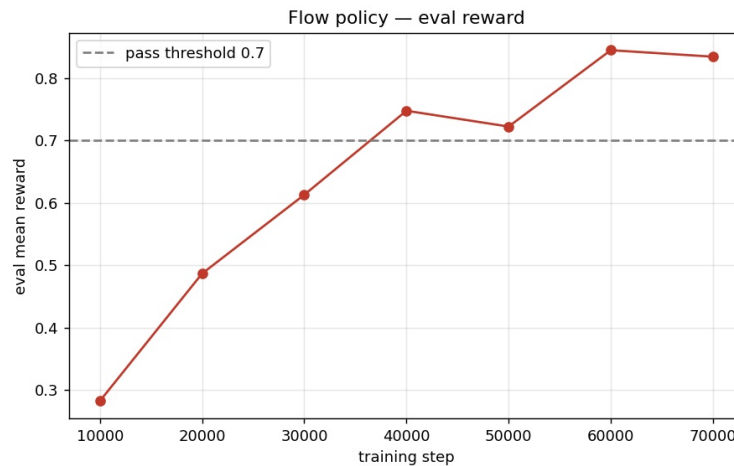
Training (eq. 2): sample $A_{t,0} \sim \mathcal{N}(0, I)$, $\tau \sim \mathcal{U}(0, 1)$; interpolate $A_{t,\tau} = \tau A_t + (1 - \tau) A_{t,0}$; regress v_θ onto $A_t - A_{t,0}$ with MSE. **Inference**

(eq. 3): start from noise, Euler-integrate $\frac{dA}{d\tau} = v_\theta$ from $\tau = 0$ to 1 in $n = 10$ steps; $A_{t,1}$ is executed.

Training curves

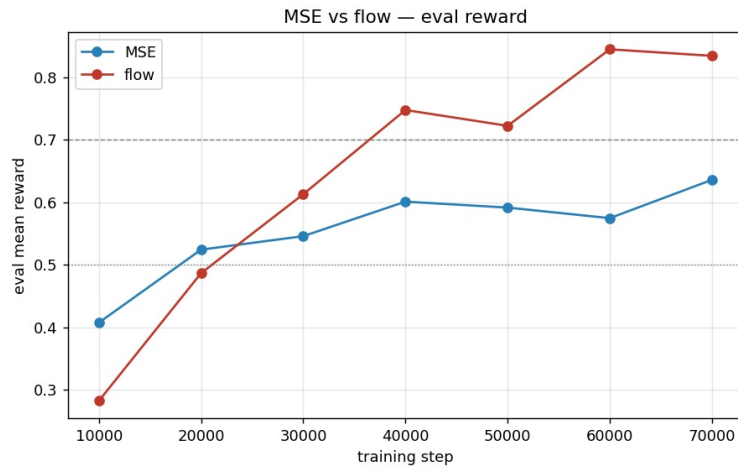


Flow training loss



Flow eval reward

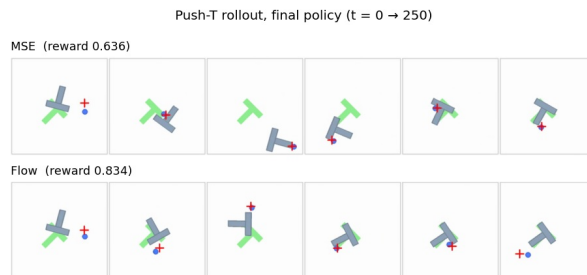
MSE vs. flow



MSE vs flow reward

Flow tracks MSE early, then pulls ahead after step 30k, ending ~0.21 reward higher.

Qualitative behavior (from rollout videos)



Rollout comparison

Reading each row left→right as the episode advances ($t = 0 \rightarrow 250$):

- **MSE** makes large, **tumbling** corrections — the one-shot chunk is the *average* of the expert’s multimodal pushes, so the block over-rotates and flings off-axis before being clawed back. It reaches the target but tends to settle **rotationally misaligned**.
- **Flow** drives the block toward the silhouette **more decisively, with smoother agent motion** — sampling a fresh chunk from noise lets it commit to one coherent push rather than averaging several. On its best episodes it locks the T tightly onto the green.
- Both retain **residual end-of-episode drift** on harder episodes (e.g. flow ep1 slides past the goal late). The flow advantage is not “never wobbles” — it is cleaner, more consistent driving onto the target, which lifts *average* coverage by +0.21.

This matches the §3 motivation: MSE regresses to the mean of a multimodal chunk distribution, whereas flow matching samples from it and so produces sharper, more committed actions.